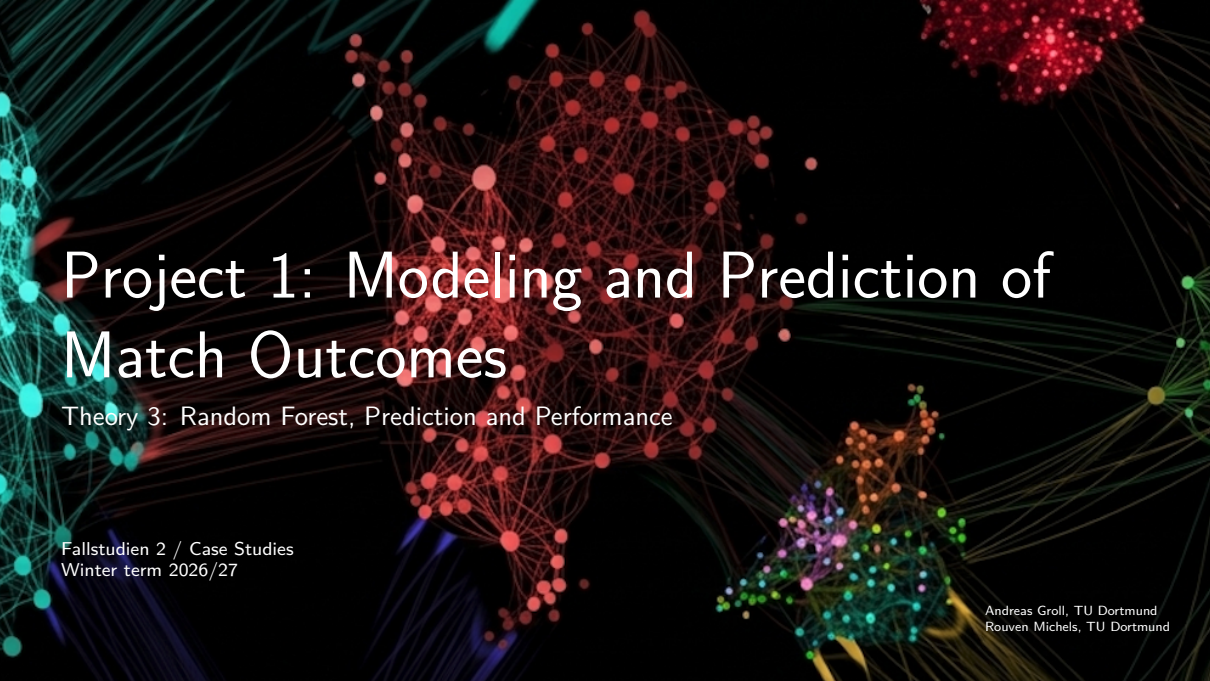


Project 1: Modeling and Prediction of Match Outcomes

Theory 3: Random Forest, Prediction and Performance

Fallstudien 2 / Case Studies
Winter term 2026/27

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Acknowledgement

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Introduction

Introduction

- the **Random Forest** as a tree-based **parallel ensemble** leaner is introduced, which is powerful w.r.t. prediction
- certain **prediction strategies** are explained
- several **performance measures** are provided
- additional aspects such as **betting strategies**, **forecasting competitions** and the **simulation** of full tournaments are briefly discussed

Random Forest

Random Forests

- a **parallel ensemble method** introduced by Breiman (2001)
- **principle**: aggregation of (large) number of **classification / regression trees**

  $\Longrightarrow$ can be used both for classification & regression purposes
- single trees are grown independently
- **final predictions**: single tree predictions are aggregated, either by majority vote (classification) or by averaging (regression)
- as single trees have high variance they are usually pruned (lowers variance but increases bias)
- main goal of Random Forest: decrease variance

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- main goal of Random Forest: decrease variance $\Longrightarrow$ decrease correlation between single trees.

Random Forests

$\Longrightarrow$ RF incorporate 2 different **randomization steps**:

1) trees are not applied to the original sample but to **bootstrap samples** or random subsamples of the data.

2) at each node a **(random) subset of the predictors** of size `mtry` is drawn that are used to find the best split.

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By de-correlating and combining many trees $\Longrightarrow$ predictions with **low bias and reduced variance**.

Main drawback: **loss in interpretation**, as final model is rather a black box.

Random Forests - Example: Football Data

- response: metric variable ***Number of Goals***
- predefined number of trees B (e.g., $B = 5000$) is fitted based on (bootstrap samples of) the training data
- prediction of new observation: covariate values are dropped down each of the regression trees, resulting in B predictions $\Longrightarrow$ average
- use predicted expected value as event rate $\hat{\lambda}$ of a Poisson distribution $Po(\lambda)$

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- 2 slightly different variants:

  1) **classical RF** algorithm (Breiman, 2001) from `R`-package `ranger` (Wright and Ziegler, 2017)

  2) RFs based on **conditional inference trees**: `cforest` from `R`-package `party` (Hothorn et al., 2006)

Interpretation of a Random Forest Model

- Principally, due to the aggregation of many trees, the RF is hard to interpret!
- The relevance of single covariates can be assessed by the concept of **Variable Importance (VI)**.
- Different variants exist:
  - **Permutation-based VI** measures how much a model's prediction performance (on the out-of-bag observations) decreases when a variable's values are randomly shuffled.
  - **Gini importance** measures the total reduction of the Gini impurity of the data set when a particular feature is used for splitting.

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Additional insight can be obtained by employing so-called **Interpretable Machine Learning (IML)** techniques. A nice overview of these can be found in Molnar (2020).

Random Forest Variable Importance

```
library(party)
form.rf <- as.formula(paste("Goals ~ ",paste(colnames(Euro_train)[6:10],collapse="+")))
rf.obj <- cforest(form.rf, data = Euro_train)

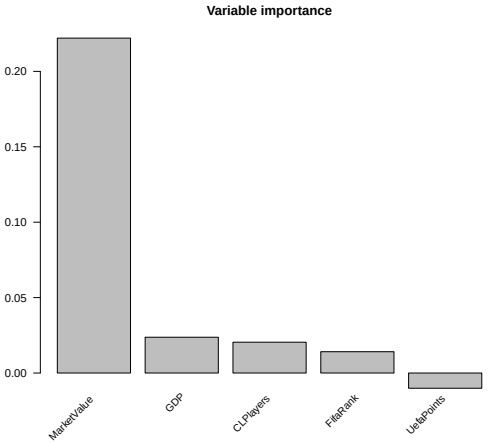
imp <- sort(varimp(rf.obj), decreasing = TRUE)
bp <- barplot(imp, main = "Variable importance", axes = FALSE, axisnames = FALSE)
text(bp, par("usr")[3], labels = names(imp), srt = 45,
     adj = c(1.15, 1.15), xpd = TRUE, cex = 0.9)
axis(2, las = 1)
```

Random Forest Variable Importance

```
library(
form.rf
rf.obj <

imp <- s
bp <- ba
text(bp,
     adj
axis(2,
```

```
+")))
```



Variable importance of a random forest fitted to the UEFA EURO 2004 – 2020 data. Number of goals is used as response, 5 features as described in Chapter 1 are used as predictors.

Prediction Strategies

Prediction Strategies

- As we know from the previous chapter, a model's **ability to generalize** should be evaluated on new (unseen) **test data**.
- Particularly, if we want to compare several different statistical or machine learning models w.r.t. their predictive performance, we need to set up some type of **prediction competition**.
- In sports analytics, often the prediction of new, upcoming matches is of interest anyway.

In the following, we will illustrate certain prediction strategies using the example of the UEFA EURO football data.

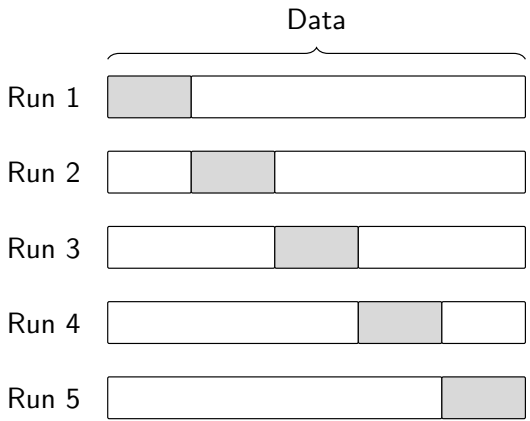
K-fold CV

- Generally, for comparing different models, often **K-fold cross validation** is used.
- In K-fold CV, the data is split into K equally sized folds, which are successively used for training and testing.
- In each CV-iteration, $K - 1$ folds are used for training, and the left out fold for testing.
- Typical choices for K are: $K \in \{5, 10, N\}$.

K-fold CV

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Scheme of 5-fold CV

K-fold CV

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- Typical choices for K are: $K \in \{5, 10, N\}$.
- However, standard K-fold CV not suitable for data with a certain **time structure**.

K-fold CV

Example: UEFA EURO 2004-2024 football data

For sports data, it can be reasonable to make some adjustments and **account for the specific structure** of the data. For football **tournament data** one could e.g. proceed as follows:

1. Form a training data set containing 5 out of 6 EUROs.
2. Fit each of the methods to the current training data (including tuning!!!).
3. Predict the left-out EURO with each method.
4. Iterate steps 1-3 such that each EURO is once the left-out one.
5. Compare predicted and real outcomes via suitable performance measures.

For step 5, we need suitable **performance measures**.

Rolling Window

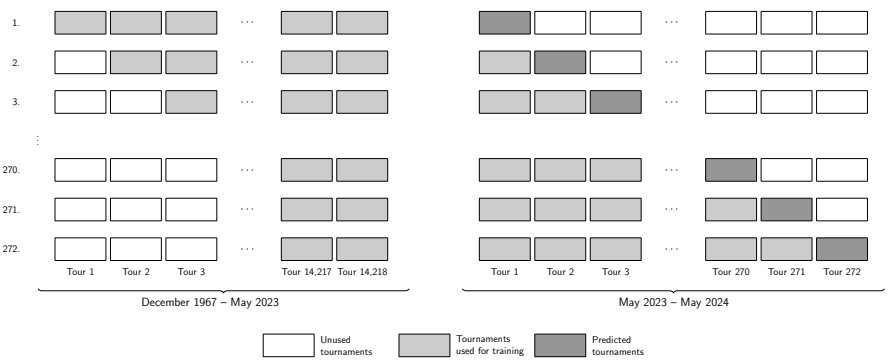
Example: ATP Grand Slam tennis tournament data

1. Construct training data using all matches of a certain number of historical tournaments.
2. Predict all matches of the next tournament; save predictions and actual results for later performance evaluation.
3. After predicting one tournament, update the training data by replacing the oldest tournament with the next one (in chronological order).
4. Repeat the process for all tournaments, progressively incorporating previously predicted tournaments into the training data.
5. Compare predicted and real outcomes via suitable performance measures.

Rolling Window

Example: ATP Grand Slam tennis tournament data

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Scheme of rolling window strategy

Rolling Window

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Alternatively, an **expanding window** approach could be used, where the training data is successively increased by the next tournament (but no old tournament drops out).

Performance Measures

Prediction of Goals – Example: Football Data

Assume there is a test data set consisting of N matches, which we want to predict.

- let now $y_{k\ell i}$, for $k, \ell = 1, \ldots, m$ and $i \in \{1, \ldots, N\}$, denote the observed number of goals scored by team k against team ℓ in match i
- $\hat{y}_{k\ell i}$ the corresponding predicted value
- 2 **quadratic errors**: $(y_{k\ell i} - \hat{y}_{k\ell i})^2$ and $((y_{k\ell i} - y_{\ell k i}) - (\hat{y}_{k\ell i} - \hat{y}_{\ell k i}))^2$
- 2 **absolute errors**: $|y_{k\ell i} - \hat{y}_{k\ell i}|$ and $|(y_{k\ell i} - y_{\ell k i}) - (\hat{y}_{k\ell i} - \hat{y}_{\ell k i})|$

Prediction of Match Outcomes – Example: Football Data

- True ordinal match outcomes: $\tilde{y}_1, \ldots, \tilde{y}_N$ with $\tilde{y}_i \in \{1, 2, 3\}$, for all (test) matches N.
- Predicted probabilities $\hat{\pi}_{1i}, \hat{\pi}_{2i}, \hat{\pi}_{3i}, \ i = 1, \ldots, N$.
- Let G_{1i} and G_{2i} denote the goals scored by 2 competing teams in match i

  $\Longrightarrow$ compute $\hat{\pi}_{1i} = P(G_{1i} > G_{2i}), \hat{\pi}_{2i} = P(G_{1i} = G_{2i})$ and $\hat{\pi}_{3i} = P(G_{1i} < G_{2i})$ based on the corresponding Poisson distributions $G_{1i} \sim Po(\hat{\lambda}_{1i})$ and $G_{2i} \sim Po(\hat{\lambda}_{2i})$ with estimates $\hat{\lambda}_{1i}$ and $\hat{\lambda}_{2i}$ (via Skellam distribution).
- **Benchmark**: **bookmakers** $\Longrightarrow$ compute the 3 quantities $\tilde{\pi}_{ri} = 1/\text{odds}_r$, $r \in \{1, 2, 3\}$, normalize with $c_i := \sum_{r=1}^{3} \tilde{\pi}_{ri}$ (adjust for bookmakers' margins)

  $\Longrightarrow$ estimated probabilities $\hat{\pi}_{ri} = \tilde{\pi}_{ri}/c_i$

Performance measures – Example: Football Data

(a) **Multinomial likelihood**: probability of correct prediction; for a single match defined as

$$\hat{\pi}_{1i}^{\delta_{1\tilde{y}_i}} \hat{\pi}_{2i}^{\delta_{2\tilde{y}_i}} \hat{\pi}_{3i}^{\delta_{3\tilde{y}_i}},$$

with $\delta_{ri} := 1$, if $r = i$, and $\delta_{ri} := 0$ else (Kronecker's delta).

(b) **Classification rate**: checks if match i is correctly classified using the indicator function

$$\mathbb{1}(\tilde{y}_i = \underset{r \in \{1,2,3\}}{\arg\max} \, (\hat{\pi}_{ri})) .$$

(c) **Rank probability score (RPS)**: explicitly accounts for the ordinal structure and is defined as

$$\frac{1}{3-1} \sum_{r=1}^{3-1} \left(\sum_{l=1}^{r} \hat{\pi}_{li} - \delta_{l\tilde{y}_i} \right)^2 .$$

Further Aspects

Betting Strategies – Example: Football Data

In addition to classic statistical performance measures, in sports analytics also certain **betting strategies** could be employed. Particularly, in football the following strategies are common:

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For every match i and each of the possible three outcomes $r \in \{1, 2, 3\}$ calculate expected return:

$$E[return_{ri}] = \hat{\pi}_{ri} * odds_{ri} - 1 \,.$$

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$$E[return_{ri}] = \hat{\pi}_{ri} * odds_{ri} - 1\,.$$

Choose outcome with highest expected return and only place bet if expected return is positive:

$$\max_{r \in \{1,2,3\}} E[return_{ri}] > \tau = 0\,.$$

Betting Strategies – Example: Football Data

Extensions:

Koopman and Lit (2015): use different thresholds $\tau > 0 \Longrightarrow$ overall mean return could be increased.

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Boshnakov et al. (2017): use varying stake sizes based on the Kelly criterion (Kelly, 1956). $\implies$ determines optimal stake for single bets in order to maximize the return considering size of the odds and the winning probability.

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Important notes:

- We advise against placing real bets recklessly using statistical models, as those have to be very good to achieve positive returns, as the bookmakers include **substantial margins** in their odds.
- Nevertheless, one can challenge the models using the bookmakers' odds and these betting strategies for **ex-post analyses** via **hypothetical** betting.

Further (fun) competitions – Example: Football Data

Specifically in male professional football there are also certain additional (fun) competitions where the predictions of a statistical model can be useful:

- the German online tipping platform **Kicktipp** (see `https://www.kicktipp.de/`),
- public forecast competitions, such as `fifaexperts.com` in 2018,
- forecast competitions **from academia** (see, e.g., the UEFA EURO 2020 prediction competition set up by Prof. Dr. Prof. Claus Ekstrøm on github, see `https://www.sandsynligvis.dk/2021/06/02/uefa-euro-2020-prediction-competition/`).

Further (fun) competitions – Example: Football Data

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Pos	+/-	Name	SPA ENG 2:1	P	B	S	G
1.		DrDaniel	2:1$_4$	4	20	1,08	100
2.		Jessica	2:1$_4$	4	17	1,08	94
3.	▲2	Maike	2:1$_4$	4	17	0,58	90
4.	▲2	Caro	2:1$_4$	4	20	0,08	90
5.	▼2	Der_Libero	1:1	0	17	1,00	89
6.	▲3	Vivi	2:1$_4$	4	20	1,08	88
7.	▼1	Simon.K	3:1$_2$	2	20		88
8.	▼4	JonasH	1:2	0	15	1,00	87
9.	▼1	TimTippt	0:1	0	12		85
10.		CombinedMLModel	0:1	0	17		84
10.		joris		0	17		84
12.		TLP	1:2	0	14		83
13.		Johanna		0	14	1,00	79
14.	▲2	Inken	2:1$_4$	4	17	0,08	79
15.	▼1	Erik		0	19		79

Final standing in UEFA EURO 2024 **Kicktipp** league (with 42 colleagues)

Further (fun) competitions – Example: Football Data

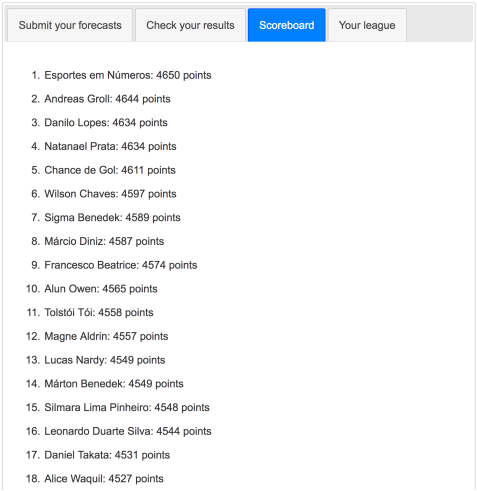
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Final standing in FIFA 2018 forecast competition **fifaexperts.com** ($>$ 500 participants)

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Further (fun) competitions – Example: Football Data

Specifical … where the prediction …

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- publ…
- forec… petition set u… //ww… on/).

	TRPS
Brandt (ELO)	0.0942
Current strength	0.1011
Mads	0.1025
Random forest	0.1045
Podlewski	0.1054
Bookmaker consensus	0.1066
XGBoost	0.1089
FK	0.1319
CD	0.1478
Simple	0.1667

Final standing in **UEFA 2020 forecast competition** from Prof. Claus Ekstrøm

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Simulation and Prediction of Tournaments

Example: Football Data

Based on the model estimates on a suitable training data set, **individual new matches** and, hence, also **whole tournaments** can be simulated.

It is important, to carefully follow the respective **tournament course** and **rules**. Moreover, in the knockout stage one has to account for **extra time** and **penalty shootouts**.

Finally, one obtains **survival probabilities** for each participating team and tournament round.

Simulation and Prediction of Tournaments

Example:

Based on , hence, also **who**

It is impo in the knockout

Finally, o d.

		Round of 16	Quarter finals	Semi finals	Final	European Champion	Bookmakers consensus
1.	FRA	93.6	72.6	50.2	31.2	19.2	17.6
2.	ENG	97.0	72.2	47.8	28.8	16.7	20.5
3.	GER	95.2	65.7	40.8	24.6	13.7	14.0
4.	ESP	93.5	66.6	38.4	21.6	11.4	9.7
5.	POR	95.9	60.4	37.1	20.4	10.8	10.1
6.	NED	82.1	55.3	31.6	16.1	7.6	4.9
7.	ITA	86.8	54.0	26.1	12.6	5.6	4.9
8.	BEL	93.8	51.6	24.5	11.2	4.9	4.5
9.	CRO	71.9	37.6	15.0	5.9	2.1	2.2
10.	DEN	75.5	35.2	15.0	6.0	2.0	2.0
11.	SUI	75.3	33.4	12.9	4.9	1.6	1.2
12.	AUT	53.4	26.0	10.6	3.6	1.1	1.1
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮

Winning and survival probabilities UEFA EURO 2024